

Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 2: Optoelectronics

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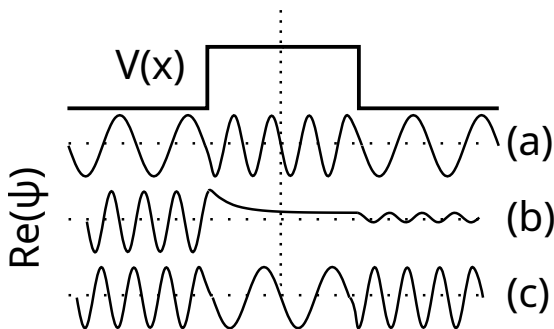
M2U2 Plan

- 1 p-n junction diode – forward and reverse conditions
- 2 Solar cell
- 3 Photo-diode
- 4 Direct and indirect band gap semiconductors
- 5 Light Emitting Diode
- 6 Tunneling diode

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Quantum mechanical tunneling



Applications

- Tunneling diode
- Nobel prize in Physics 2025
- Scanning Tunneling Microscope
- Frustrated total internal reflection
- α decay – Nuclear fission
- Everything on earth – Nuclear fusion

Tunneling – The reason for everything

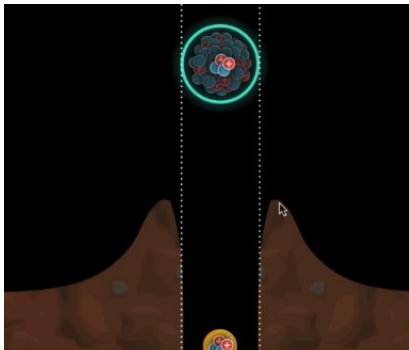
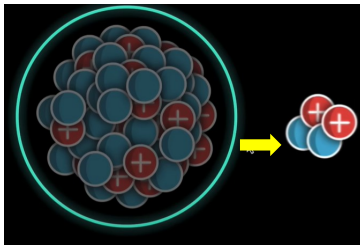
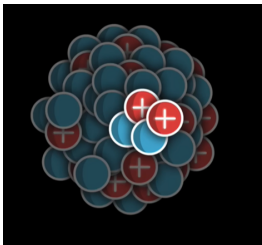
Key Insight



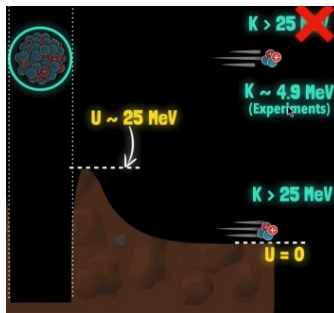
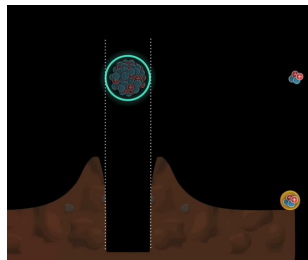
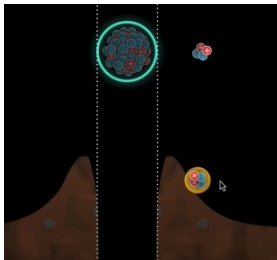
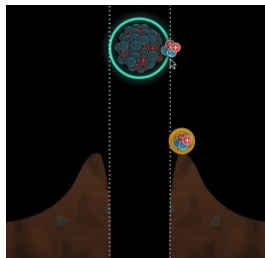
Frustrated total internal reflection



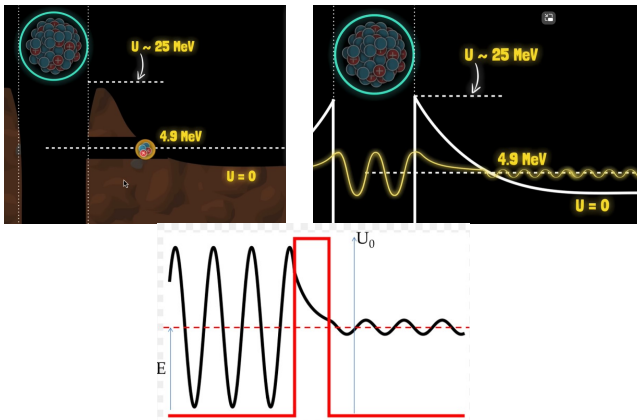
α decay – Nuclear fission



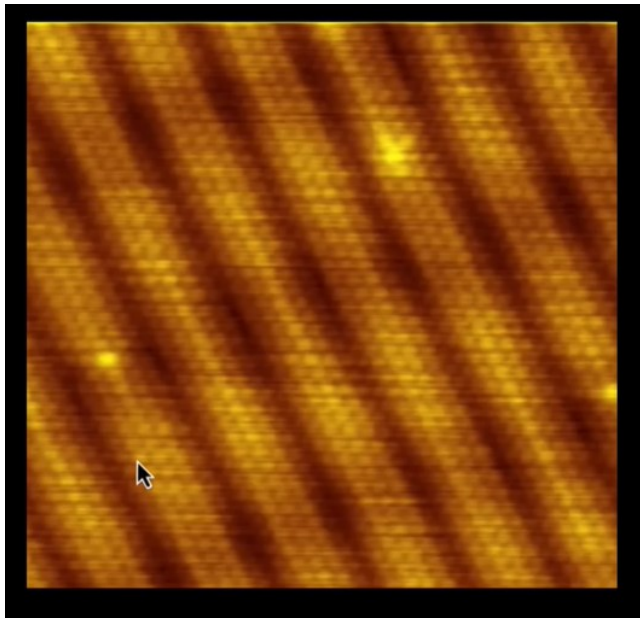
α decay – Nuclear fission



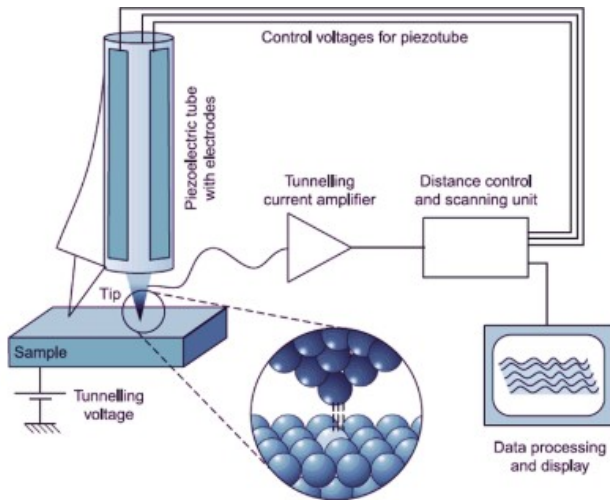
α decay – Nuclear fission



Scanning Tunneling Microscope (STM)

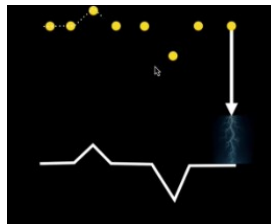
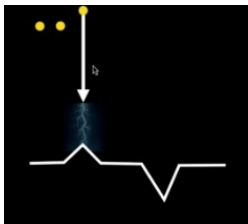
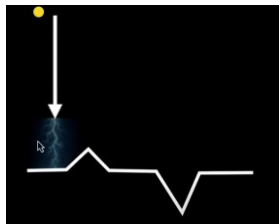


STM – Construction



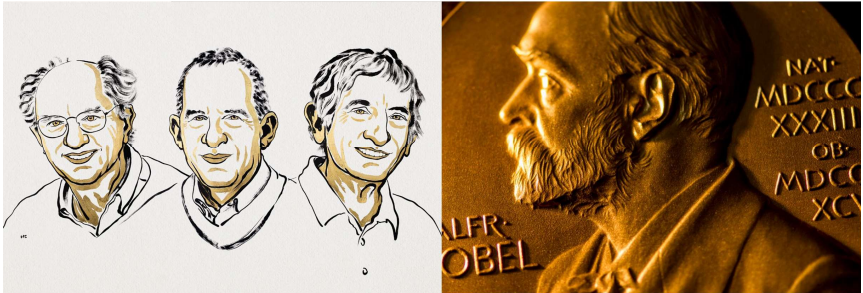
Quantum tunneling and how STM works? [[Click here for video](#)]

STM – Working principle



A Boy and his Atom [[Click here for video](#)]

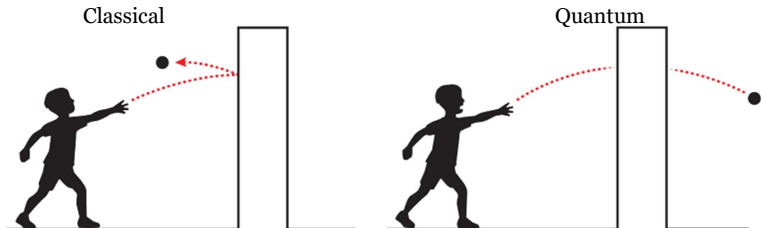
Nobel Prize in Physics 2025



Prof. Clarke, Devoret, Martinis

“for the discovery of macroscopic quantum mechanical tunneling
and
energy quantization in an electric circuit”

Quantum tunneling



Tunneling From Nobel Prize website

Tunneling

- Classically forbidden
- Quantum mechanically allowed
- Many examples of quantum tunneling at microscopic scale
 - Nuclear fission
 - Thermionic emission

Microscopic tunneling



Nuclear fission

Macroscopic tunneling



Electrons in a conductor



Cooper pairs
in a superconductor/insulator/superconductor sandwich



Shared/Macroscopic wavefunction leads to tunneling